9.1 Introduction

Windings are used in the electrical field such as motors, generators, transformers, loud speakers and in all kinds of electrical measuring instruments. The main objective of the use of winding is to produce the magnetic field according to the capacity of the electrical machines. In this lesson, we have to learn about the materials needed for making the windings and different types of DC and AC windings.

9.2 Winding materials

Generally two types of important materials are used in the process of winding the motors and generators. They are,

i. Electric conductors
ii. Electrical insulating materials.

i. Electric conductors

Materials having low resistance allows the electric current to flow easily through conductors are called as electric conductors. Generally, it is made up of metals. For winding, copper conductors are mainly used for making coils and aluminium conductors are used next.

a. Types of conductors

For the following windings are the types of conductors commonly used.

i. Cotton covered winding wires.
ii. Silk covered winding wires.
iii. Paper covered winding wires.
iv. Varnished glass paper covered winding wires.
v. Enamel coated round shaped winding wires.

Enamel coated round wires are mainly used in small and medium type of motors having semi-closed slots while rectangular conductors are used in larger types of motors having open type slots.

ii. Electrical insulating materials

Materials having very high resistance and also having property not to permit the electric current through them are called insulating materials.

a. Varnishes

It helps the winding to improve the insulation resistance, dielectric strength and mechanical strength. It is used to protect the windings from dust and moisture. It is also used to give coating on the moisture absorbing materials.

9.3 Wire gauge plate

9.3.1 Methods of representing the size of the winding wires

The sizes of the conductors may be notified in two ways as follows,

i. By SWG numbers – a steel "Wire Gauge Plate" is used
ii. By its diameter in millimeter by metric method – An instrument micrometer (Screw gauge) is used.

To Find gauge number of a conductor by using standard wire gauge plate

The sizes and the gauge of the wires can be found by a device called standard gauge plate, which consists of a circular steel plate having notches of different widths around their edges. The diameter of the plate will be generally in 3 ¾ inches (95mm) with thirty six notches. Each notch is stamped with a number. The size

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